

Theorem 2.17 [36] *A connected graph G with n vertices has $rc(G) < \frac{20n}{\delta(G)}$.*

The proof of Theorem 2.17 uses the concept of connected two-step dominating set. Krivelevich and Yuster first proved that for a connected graph H with minimum degree k and n vertices, there exists a two-step dominating set S whose size is at most $\frac{n}{k+1}$ and there is a connected two-step dominating set S' containing S with $|S'| \leq 5|S| - 4$. They found two edge-disjoint spanning subgraphs in a graph G with minimum degree at least $\lfloor \frac{\delta-1}{2} \rfloor$. Then they derived a rainbow coloring for G by giving a rainbow coloring to each subgraphs according to its connected two-step dominating set.

The authors noted that the constant 20 obtained by their proof is not optimal and can be slightly improved with additional effort. However, from the example below Theorem 2.15, one cannot expect to replace C by a constant smaller than 3.

Motivated by the results of Theorems 2.14, 2.15 and 2.17, Schiermeyer raised the following open problem in [53].

Problem 2.18 [53] *For every $k \geq 2$ find a minimal constant c_k with $0 < c_k \leq 1$ such that $rc(G) \leq c_k n$ for all graphs G with minimum degree $\delta(G) \geq k$. Is it true that $c_k = \frac{3}{k+1}$ for all $k \geq 2$?*

This is true for $k = 2, 3$ as shown before ($c_2 = 1$ and $c_3 = \frac{3}{4}$).

Recently, Chandran, Das, Rajendraprasad and Varma [11] nearly settled the above problem. They used the concept of a connected two-way two-step dominating set in the argument and they first proved the following result.

Theorem 2.19 [11] *If D is a connected two-way two-step dominating set in a graph G , then $rc(G) \leq rc(G[D]) + 6$.*

Furthermore, they gave a nearly sharp bound for the size of D by showing that every connected graph G of order $n \geq 4$ and minimum degree δ has a connected two-way two-step dominating set D of size at most $\frac{3n}{\delta+1} - 2$; moreover, for every $\delta \geq 2$, there exist infinitely many connected graphs G such that $\gamma_c^2(G) \geq \frac{3(n-2)}{\delta+1} - 4$. Then the following result is easy.

Theorem 2.20 [11] *For every connected graph G of order n and minimum degree δ ,*

$$rc(G) \leq \frac{3n}{\delta+1} + 3.$$

Moreover, for every $\delta \geq 2$, there exist infinitely many connected graphs G such that $rc(G) \geq \frac{3(n-2)}{\delta+1} - 1$.

Theorem 2.20 answers Problem 2.18 in the affirmative but up to an additive constant of 3. Moreover, this bound is seen to be tight up to additive factors by the construction

mentioned in [9] (see the example below Theorem 2.15) and [23]. And therefore, for graphs with linear minimum degree ϵn , the rainbow connection number is bounded by a constant.

Recently, Dong and Li [22] derived an upper bound on rainbow connection numbers of graphs under given degree sum condition σ_2 . Recall that for a graph G , $\sigma_2(G) = \min\{d(u) + d(v) \mid u, v \text{ are independent in } G\}$. Clearly, the degree sum condition σ_2 is weaker than the minimum degree condition.

Theorem 2.21 [22] *For a connected graph G of order n , $rc(G) \leq 6\frac{n-2}{\sigma_2+2} + 7$.*

Similar to the method of Theorem 2.20, they derived that every connected graph G of order n with at most one pendant vertex has a connected two-way two-step dominating set D of size at most $6\frac{n-2}{\sigma_2+2} + 2$. Then by using Theorem 2.19, they got the theorem.

From the example below Theorem 2.15, we know their bound are seen to be tight up to additive factors. Note that by the definition of σ_2 , we know $\sigma_2 \geq 2\delta$, so from Theorem 2.21, we can derive $rc(G) \leq 6\frac{n-2}{\sigma_2+2} + 7 \leq \frac{3(n-2)}{\delta+1} + 7$. And the bound in Theorem 2.21 can be seen as an improvement of that in Theorem 2.20.

With respect to the relation between $rc(G)$ and the connectivity $\kappa(G)$, mentioned in [53], Broersma asked a question at the IWOCA workshop:

Problem 2.22 [53] *What happens with the value $rc(G)$ for graphs with higher connectivity?*

For $\kappa(G) = 1$, Theorem 2.14 means that if G is a graph of order n , connectivity $\kappa(G) = 1$ and $\delta \geq 3$. Then $rc(G) \leq \frac{3n-1}{4}$. For $\kappa(G) = 2$, in the proof of Theorem 2.12, as we mentioned above, Caro, Lev, Roditty, Tuza and Yuster derived:

Theorem 2.23 [9] *If G is a 2-connected graph with n vertices then $rc(G) \leq \frac{2n}{3}$.*

That is, if G is a graph of order n , connectivity $\kappa(G) = 2$. Then $rc(G) \leq \frac{2n}{3}$.

From Theorem 2.20, we can easily obtain an upper bound of the rainbow connection number according to the connectivity:

$$rc(G) \leq \frac{3n}{\delta+1} + 3 \leq \frac{3n}{\kappa+1} + 3.$$

Therefore, for $\kappa(G) = 3$, $rc(G) \leq \frac{3n}{4} + 3$; for $\kappa(G) = 4$, $rc(G) \leq \frac{3n}{5} + 3$. Motivated by the results in [9], and by using the Fan Lemma, Li and Shi [41] improved this bound by showing the following result.

Theorem 2.24 ([41]) *If G is a 3-connected graph with n vertices, then $rc(G) \leq \frac{3(n+1)}{5}$.*

However, for general connectivity, there is no upper bound which is better than $\frac{3n}{\kappa+1} + 3$.

The following result is an important ingredient in the proof of Theorem 2.12 in [9].

Theorem 2.25 [9] *If G is a connected bridgeless graph with n vertices, then $rc(G) \leq \frac{4n}{5} - 1$.*

From Theorem 2.20, we can also easily obtain an upper bound of the rainbow connection number according to the edge-connectivity λ :

$$rc(G) \leq \frac{3n}{\delta + 1} + 3 \leq \frac{3n}{\lambda + 1} + 3.$$

Note that all the above upper bounds are determined by n and other parameters such as (edge)-connectivity, minimum degree. Diameter of a graph, and hence its radius, are obvious lower bounds for rainbow connection number. Hence it is interesting to see if there is an upper bound which is a function of the radius r or diameter alone. Such upper bounds were shown for some special graph classes in [11] which we will introduce in the sequel. But, for a general graph, the rainbow connection number cannot be upper bounded by a function of r alone. For instance, the star $K_{1,n}$ has a radius 1 but rainbow connection number n . Still, the question of whether such an upper bound exists for graphs with higher connectivity remains. Basavaraju, Chandran, Rajendraprasad and Ramaswamy [4] answered this question in the affirmative. The key of their argument is the following lemma, and in the proof of this lemma, we can obtain a connected $(k - 1)$ -step dominating set from a connected k -step dominating set.

Lemma 2.26 [4] *If G is a bridgeless graph, then for every connected k -step dominating set D^k of G , $k \geq 1$, there exists a connected $(k - 1)$ -step dominating set $D^{k-1} \supset D^k$ such that*

$$rc(G[D^{k-1}]) \leq rc(G[D^k]) + \min\{2k + 1, \zeta\},$$

where $\zeta = iso(G)$.

Given a graph G and a set $D \subset V(G)$, A D -ear is a path $P = (x_0, x_1, \dots, x_p)$ in G such that $P \cap D = \{x_0, x_p\}$. P may be a closed path, in which case $x_0 = x_p$. Further, P is called an *acceptable* D -ear if either P is a shortest D -ear containing (x_0, x_1) or P is a shortest D -ear containing (x_{p-1}, x_p) . Let $\mathcal{A} = \{a_1, a_2, \dots\}$ and $\mathcal{B} = \{b_1, b_2, \dots\}$ be two pools of colors, none of which are used to color $G[D^k]$. A D^k -ear $P = (x_0, x_1, \dots, x_p)$ will be called *evenly colored* if its edges are colored $a_1, a_2, \dots, a_{\lceil \frac{p}{2} \rceil}, b_{\lfloor \frac{p}{2} \rfloor}, \dots, b_2, b_1$ in that order. Basavaraju, Chandran, Rajendraprasad and Ramaswamy proved this lemma by constructing a sequence of sets $D^k = D_0 \subset D_1 \subset \dots \subset D_t = D^{k-1}$ and coloring the new edges in every induced graph $G[D_i]$ such that the following property is maintained for all $0 \leq i \leq t$: every $x \in D_i \setminus D^k$ lies in an evenly colored acceptable D^k -ear in $G[D_i]$.

The following theorem can be derived from Lemma 2.26 easily.

Theorem 2.27 [4] *For every connected bridgeless graph G ,*

$$rc(G) \leq \sum_{i=1}^r \min\{2i + 1, \zeta\} \leq r\zeta,$$

where r is the radius of G .